Exact Rational Stream-Dual Certificates for a Two-Bidder, Two-Item Auction

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Abstract

We study revenue maximization for two additive bidders and two heterogeneous items when all four values are independent and uniform on $[0, 1]$. The mechanism class contains every measurable, integrable, possibly randomized mechanism that is dominant-strategy incentive compatible, ex-post individually rational, and ex-post feasible. We construct an explicit rational continuous weak-duality witness in a stream-function representation. Its divergence-free, tangent-boundary correction preserves the revenue identity, while itemwise feasibility reduces the resulting objective to the integral of a pointwise maximum of polynomial virtual coefficients. A deterministic adaptive tensor-Bernstein certificate upper-bounds this continuous envelope with directed fixed-point rounding and exact integer accumulation. The resulting bound is

$$\mathrm{OPT} \le \frac{930318295428931}{1048576000000000} = 0.887220664433413505554199921875.$$

This is strictly below the previously reported continuous DSIC certificate 0.8919. We also give an explicit deterministic mechanism with exact revenue

$$\frac{26232788323031183}{30000000000000000} = 0.8744262774343727\ldots.$$

The endpoints do not match, so the unrestricted optimum remains open. The contribution is the instance-specific rational witness and its independently reproducible exact continuous certification, not a new general theory of continuous auction duality.

Scope of the result. The unrestricted continuous DSIC optimum is not determined here. We prove a fully specified deterministic lower bound and an independent exact global upper bound for all randomized mechanisms. The values do not match.

1 Introduction

Optimal auction design is completely understood for a single item under the classical regularity conditions of Myerson [6]. With several items, bidder types become multidimensional and dominant-strategy incentive compatibility imposes global cross-type restrictions. Even the canonical case of two additive bidders, two items, and independent uniform values has resisted an exact characterization [4, 8].

Computational work has produced strong feasible mechanisms and rigorous upper bounds. Affine-maximizer searches provide transparent deterministic benchmarks [8], while GemNet constructs exactly strategy-proof menu mechanisms after a compatibility transformation [9]. On the dual side, Jiang, Parkes, and Wang developed flow-conserving DSIC kernels and a continuous lifting theorem; their reported 50×50 uniform certificate is 0.8919 for the present instance [4]. These results leave room for a smaller certificate whose arithmetic and continuous integration can be replayed exactly.

We pursue a target-specific continuous witness rather than another grid lift. Pointwise DSIC represents each bidder's truthful utility as a convex potential [7]. A radial integration identity produces a base vector field with the required distributional source. We then add a rational polynomial curl generated by a stream function. The correction has zero divergence and zero normal boundary component, so it changes neither the conservation law nor the revenue pairing. Ex-post feasibility bounds the paired allocation pointwise by the positive maximum of the two bidders' field components.

The remaining problem is a four-dimensional continuous polynomial-envelope integral. We certify it using tensor Bernstein control coefficients, exact rational construction, directed fixed-point rounding, and a certificate-aware adaptive dyadic traversal concentrated near unresolved winner-switching boxes. The certificate proves the exact upper bound $\frac{930318295428931}{1048576000000000}$. A separate implementation reconstructs the polynomial charts and repeats the complete base-depth-20 plus adaptive-depth-one traversal without importing the formal verifier. As a secondary result, exact rational polytope integration and a menu-surcharge calculation certify the lower endpoint $\frac{262327883230311183}{300000000000000000}$.

Contributions. The paper makes three bounded contributions:

1. an explicit 32-parameter rational stream correction that yields a valid continuous DSIC weak-duality witness for the canonical two-bidder, two-item uniform instance;
2. an exact Bernstein-based integration certificate, including directed rounding, overflow bounds, deterministic transcripts, and an independent full-depth replay; and
3. the rigorous bracket

$$\frac{262327883230311183}{300000000000000000} \leq \mathrm{OPT} \leq \frac{930318295428931}{1048576000000000}.$$

We do not claim a general new continuous duality theorem, a one-to-one identification with every flow kernel, or global optimality of the lower mechanism.

2 Relation to prior dual formulations

Rochet's cyclic-monotonicity characterization turns DSIC into convexity of truthful utility and identifies allocation vectors with subgradients [7]. Continuous duality for the one-bidder multi-item problem has been developed through transport and partial-derivative formulations [2, 3]. These results motivate integration-by-parts certificates but do not directly enforce item supply across several bidders.

Multi-bidder dual formulations also predate the present work. Finite-type flow-induced virtual values arise in unified Bayesian mechanism-design duality [1]; Zuo considers both Bayesian and dominant- strategy formulations [10]; and Beckmann-style transport gives a continuous strong-duality formulation for multi-item, multi-bidder BIC auctions in interim space [5]. The latter problem uses weaker incentive and individual-rationality requirements than the pointwise DSIC and ex-post IR requirements studied here.

Most directly, Jiang, Parkes, and Wang construct nonnegative, flow-conserving DSIC deviation kernels, derive continuous weak duality, and lift coarse uniform-grid certificates to continuous type spaces [4]. Our proof shares the same high-level weak-duality architecture: conservation eliminates payments, the dual object induces virtual coefficients, and ex-post feasibility bounds virtual surplus itemwise. The representation and certification are different. We use a signed vector field with a distributional radial source plus a divergence-free tangent stream correction, and prove its validity directly against Lipschitz convex utilities. We do not assert that every such stream field is generated by a nonnegative deviation kernel, or conversely. The novelty claimed here is narrower: an explicit rational witness for one canonical DSIC instance and an exact Bernstein certificate for its continuous objective.

3 Setting and proved bounds

Write a report profile as

$$(x, y, z, w) = (v_{11}, v_{12}, v_{21}, v_{22}) \in [0, 1]^4.$$

The four coordinates are independent and uniform. A mechanism specifies jointly measurable expected allocation probabilities $x_{ij} : [0, 1]^4 \to [0, 1]$ and expected payments $p_i : [0, 1]^4 \to \mathbb{R}$, with $p_i \in L^1([0, 1]^4)$. Allocations may arise from arbitrary internal randomization; all feasibility, DSIC, and ex-post-IR inequalities concern these expectations pointwise.

Theorem 1 (Audited bracket)**.** *For the unrestricted feasible, DSIC, ex-post-IR problem,*

$$\frac{26232788323031183}{30000000000000000} \le \text{OPT} \le U_{\text{stream}},$$

where

$$U_{\text{stream}} = \frac{930318295428931}{1048576000000000} = 0.887220664433413505554199 21875.$$

For comparison, the simpler uncorrected radial field gives the weaker analytic bound

$$\begin{aligned} U_{\text{rad}} &= \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9} \log \sqrt{3} \\ &\quad + \frac{2}{9\sqrt{3}} \left[\text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(\sqrt{3} - 2) \right] = 0.920577474278749\ldots. \end{aligned}$$

The lower endpoint is

$$0.87442627743437 27\ldots.$$

The lower bound is proved in Sections 4–6; the two upper bounds are proved in Section 7. The stream bound strictly improves the printed 0.8919 decimal reported in a 2026 preprint, by the exact rational difference

$$\frac{8919}{10000} - U_{\text{stream}} = \frac{4906638971069}{1048576000000000} > 0.$$

This comparison is to the reported decimal, not a reconstruction of the unreleased underlying certificate.

4 An exact base affine maximizer

Set

$$a = \frac{159}{250}, \qquad b = \frac{91}{100}, \qquad s = \frac{1137}{1000}.$$

For each feasible outcome A, let c_A and the affine score be as follows. The row number is also the tie-breaking priority.

id	allocation	score	c_A
0	seller keeps both	0	0
1	$1 \to 1$	$x - a$	a
2	$2 \to 1$	$y - a$	a
3	$\{1,2\} \to 1$	$x + y - b$	b
4	$1 \to 2$	$z - a$	a
5	$2 \to 2$	$w - a$	a
6	$\{1,2\} \to 2$	$z + w - b$	b
7	$1 \to 1,\ 2 \to 2$	$x + w - s$	s
8	$2 \to 1,\ 1 \to 2$	$y + z - s$	s

The base outcome $A^0(v)$ is the smallest-id maximizer of the nine scores. This defines the outcome on every equality surface.

Let

$$H_1(z,w) = \max\{0, z-a, w-a, z+w-b\}, \quad H_2(x,y) = \max\{0, x-a, y-a, x+y-b\}.$$

If $V_i(A)$ is bidder i's reported value for the items assigned in A, the base payments are

$$p_1^0 = H_1 - V_2(A^0) + c_{A^0}, \qquad p_2^0 = H_2 - V_1(A^0) + c_{A^0}.$$

Lemma 2. *The base mechanism is feasible, pointwise DSIC, and pointwise ex-post IR.*

Proof. Every listed outcome is feasible. Fix bidder 2's report. Apart from the constant H_1, bidder 1's utility from an induced outcome A, evaluated at the true type, is

$$V_1(A) + V_2(A) - c_A - H_1.$$

Truthful reporting selects a maximizer of the expression preceding H_1; no report can give greater utility. The same argument applies to bidder 2. It remains valid on ties because the selected row is still a maximizer. The outcomes excluding bidder 1 attain H_1, so the winning score is at least H_1, and bidder 1's truthful utility is nonnegative. The symmetric argument proves bidder 2's IR. □

For a bidder facing opponent report $q = (q_1, q_2)$, define

$$H(q) = \max\{0, q_1 - a, q_2 - a, q_1 + q_2 - b\}.$$

The base taxation menu has prices

$$A_0(q) = H(q) + \min\{a, s - q_2\}, \quad B_0(q) = H(q) + \min\{a, s - q_1\}, \quad C_0(q) = H(q) + b \tag{1}$$

for item 1, item 2, and the bundle, respectively; opt-out costs zero. These formulas follow directly by minimizing the outcome cost net of the opponent's value among outcomes assigning the indicated bundle.

5 The entry-surcharge mechanism

Define two disjoint closed rectangles

$$D = \left([0, \tfrac{3}{10}] \times [\tfrac{23}{40}, \tfrac{3}{5}]\right) \cup \left([\tfrac{23}{40}, \tfrac{3}{5}] \times [0, \tfrac{3}{10}]\right)$$

and the measurable surcharge

$$E(q)=\begin{cases}\frac{1}{40}, & q\in D,\\ 0, & q\notin D.\end{cases}$$

Against opponent report q, offer the four options

$$(\varnothing,0),\quad (\{1\},A_0(q)+E(q)),\quad (\{2\},B_0(q)+E(q)),\quad (\{1,2\},C_0(q)+E(q)).$$

Here is a global tie rule that makes the two bidder menus compatible. First compute the base outcome $A^0(v)$ and each base assigned bundle $B_i^0(v)$. If $B_i^0(v)=\varnothing$, assign bidder i nothing. Otherwise compute the surcharged utility of $B_i^0(v)$. Retain that bundle and charge its menu price if this utility is strictly positive; assign nothing and charge zero if it is nonpositive. Thus zero-utility ties are always resolved in favor of opt-out, while nonempty ties inherit the base smallest-outcome-id rule.

Proposition 3. *The entry-surcharge mechanism is deterministic, measurable, ex-post feasible, pointwise DSIC, and pointwise ex-post IR on the entire cube, including all menu and rectangle boundaries.*

Proof. For a fixed opponent report, $E(q)$ is added equally to every nonempty option. It therefore preserves the ranking among all nonempty bundles. The base bundle $B_i^0(v)$ is a maximizing nonempty option whenever the maximum nonempty utility is positive. The stated rule hence chooses a utility-maximizing option from a menu independent of bidder i's own report. For any true type, no misreport can produce an option whose true utility exceeds the maximum of this menu, proving DSIC. Opt-out is present and is chosen at utility zero, proving ex-post IR.

Relative to $A^0(v)$, each bidder either keeps exactly the same bundle or switches to opt-out. The new allocation is consequently obtained only by deleting bundles from a feasible base allocation, so it is feasible. All maxima, comparisons, and the indicator of the closed set D are Borel measurable. The boundary rules above are literal pointwise rules, not almost-everywhere conventions. □

6 Exact revenue

Lemma 4 (Revenue of a four-option menu)**.** *Suppose*

$$0\le A,B\le C\le 1,\qquad C\le A+B.$$

Let an additive type be uniform on $[0,1]^2$*. The expected payment from the four-option menu is*

$$\begin{aligned}r(A,B,C)&=A(1-A)(C-A)+B(1-B)(C-B)\\&\quad+C\left[(1-C+B)(1-C+A)-\frac{(A+B-C)^2}{2}\right].\end{aligned}$$

Proof. The strict item-1 region has area $(1-A)(C-A)$, and the strict item-2 region has area $(1-B)(C-B)$. The bundle region starts with the rectangle $[C-B,1]\times[C-A,1]$ and removes a right triangle of side $A+B-C$. Multiplying the three areas by their prices proves the formula. Ties have area zero for this integral; their pointwise allocation was already fixed in Proposition 3. □

If the common surcharge is e, direct expansion gives

$$r(A+e,B+e,C+e)-r(A,B,C)=eG-\frac{3C}{2}e^2-\frac{1}{2}e^3, \tag{2}$$

where

$$G = (C - A)(1 - 2A) + (C - B)(1 - 2B) + (1 - C + A)(1 - C + B) - \frac{(A + B - C)^2}{2} - C(A + B - C).$$

On the first rectangle write $q = (\rho, t)$. Since $\rho \in [0, 3/10]$ and $t \in [23/40, 3/5]$, one has

$$H(q) = 0, \qquad (A, B, C) = (s - t, a, b).$$

All hypotheses of Lemma 4 hold before and after $e = 1/40$. Substitution into (2) yields

$$\Delta(t) = -\frac{2167}{10^6} - \frac{681}{40000}t + \frac{3}{80}t^2.$$

In particular,

$$\Delta(\tfrac{23}{40}) = \frac{7073}{16000000} > 0, \qquad \Delta(\tfrac{3}{5}) = \frac{559}{500000} > 0,$$

and

$$\int_{23/40}^{3/5} \Delta(t)\, dt = \frac{6209}{320000000}.$$

There are two transposed rectangles, each has free-coordinate length 3/10, and the same calculation applies to both bidders. The total surcharge gain is

$$4 \cdot \frac{3}{10} \cdot \frac{6209}{320000000} = \frac{18627}{800000000}.$$

The base mechanism's exact expected revenue is

$$R_0 = \frac{26232089810531183}{30000000000000000}.$$

The exact verifier in `certificate/ama_lower_bound` reconstructs the $9 \cdot 4 \cdot 4 = 144$ affine/pivot cells using rational arithmetic, finds 39 full-dimensional cells, triangulates them into 16,336 rational four-simplices, checks total volume one, and integrates the affine payments. The separate surcharge verifier reconstructs the polynomial calculation above using `fractions.Fraction`. Therefore

$$R_0 + \frac{18627}{800000000} = \boxed{\frac{26232788323031183}{30000000000000000}},$$

proving the lower half of Theorem 1.

7 Exact upper bounds for all randomized mechanisms

Fix bidder i and an opponent report on a full-measure set for which the conditional expected-payment slice is integrable. Write truthful utility as $u(v)$, expected allocation as $X(v) \in [0,1]^2$, and expected payment as $p(v)$. Pointwise DSIC gives

$$u(v) \geq u(r) + (v - r) \cdot X(r) \qquad \text{for every } v, r.$$

Thus u is convex and $X(r) \in \partial u(r)$. These inequalities and $0 \leq X_j \leq 1$ also make u Lipschitz. At every differentiability point, $X = \nabla u$.

Joint measurability and the assumed integrability of the payment, together with boundedness of $v \cdot X$, make $u = v \cdot X - p$ measurable and integrable. Thus the conditional identities below apply on this full-measure set of opponent reports and may then be integrated by Fubini.

Ex-post IR gives $u(0) \geq 0$. We do not normalize the mechanism: doing so would require an unnecessary integrability check for the opponent-dependent quantity $u(0)$.

Partition the square into the two radial charts

$$v = s(1,t) \quad \text{and} \quad v = s(t,1), \qquad 0 \leq s,t \leq 1,$$

each with Jacobian s. On a fixed ray set $q = (1,t)$ or $(t,1)$ and $g(s) = u(sq)$. Convexity makes g absolutely continuous and $g'(s) = q \cdot X(sq)$ almost everywhere and $g(0) = u(0)$. Since $p(sq) = sg'(s) - g(s)$,

$$\int_0^1 p(sq)s\,ds = \int_0^1 \frac{3s^2-1}{2}\,g'(s)\,ds - \frac{u(0)}{2}.$$

The two charts together contribute the term $-u(0)$, which is nonpositive and may be dropped for an upper bound. Equivalently, with

$$\phi(s) = \frac{3s^2-1}{2s},$$

conditional expected revenue is at most the integral of $\phi(s)q \cdot X(sq)$ against the chart measure $s\,ds\,dt$.

Apply this identity to both bidders and then integrate over opponent reports. For every profile and item j, feasibility gives $X_{1j} + X_{2j} \leq 1$, with both allocations nonnegative. Hence

$$\sum_i \phi(s_i)q_{ij}X_{ij} \leq \max\{0, \phi(s_1)q_{1j}, \phi(s_2)q_{2j}\}.$$

This pointwise relaxation is valid for arbitrary expected allocations and therefore for arbitrary internal randomization.

For completeness, the remaining scalar expectation can be evaluated exactly. Under the two radial charts, a uniform type can be generated by taking S with density $2s$, an independent fair choice of which coordinate is maximal, and an independent $T \sim U[0,1]$. For one fixed item its radial coordinate therefore has distribution

$$Q \sim \frac{1}{2}\delta_1 + \frac{1}{2}U[0,1].$$

Set $Y = \max\{0, \phi(S)\}$, independently of Q. For two bidders let $a = \max(Y_1, Y_2)$ and $b = \min(Y_1, Y_2)$. Conditioning on the four atom/uniform cases for Q_1, Q_2 gives

$$\mathbb{E}_{Q_1,Q_2} \max\{Y_1Q_1, Y_2Q_2\} = \frac{3}{4}a + \frac{b^2}{6a},$$

where b^2/a is defined as zero when $a = 0$. Since the two items are symmetric,

$$U_{\text{rad}} = \frac{3}{2}\mathbb{E}[a] + \frac{1}{3}\mathbb{E}\left[\frac{b^2}{a}\right]. \tag{3}$$

Put $r = 1/\sqrt{3}$. The function ϕ is increasing from zero to one on $[r,1]$, while $Y = 0$ below r. The larger of two independent S's has density $4s^3$, hence

$$\mathbb{E}[a] = \int_r^1 4s^3\phi(s)\,ds = \frac{8}{15} + \frac{4}{45\sqrt{3}}.$$

Ordering the two S's gives the second moment as the explicit integral

$$\mathbb{E}\left[\frac{b^2}{a}\right] = 8\int_r^1 \frac{s}{\phi(s)}\left(\int_r^s t\phi(t)^2\,dt\right)ds.$$

The inner integral is

$$J(s) = \frac{9}{16}s^4 - \frac{3}{4}s^2 + \frac{1}{4}\log s + \frac{3}{16} + \frac{1}{8}\log 3,$$

so the last display equals

$$16 \int_r^1 \frac{s^2 J(s)}{3s^2 - 1}\, ds.$$

Set $x = \sqrt{3}s$. The cancellation of the $\log 3$ terms in J reduces this integral to

$$\frac{1}{3\sqrt{3}} \int_1^{\sqrt{3}} x^2(x^2 - 3)\, dx + \frac{4}{3\sqrt{3}} \int_1^{\sqrt{3}} \frac{x^2 \log x}{x^2 - 1}\, dx.$$

Use $x^2/(x^2-1) = 1 + 1/(x^2-1)$. For the only non-elementary term, the substitution $y = (x-1)/(x+1)$ has upper endpoint $d = 2 - \sqrt{3}$ and gives

$$\int_1^{\sqrt{3}} \frac{\log x}{x^2 - 1}\, dx = \frac{1}{2}\int_0^d \frac{\log(1+y) - \log(1-y)}{y}\, dy = \frac{1}{2}[\mathrm{Li}_2(d) - \mathrm{Li}_2(-d)],$$

where

$$\frac{d}{dz}\mathrm{Li}_2(z) = -\frac{\log(1-z)}{z}$$

was used in the last equality. The remaining polynomial and $\int \log x\, dx$ terms then give

$$-\frac{26}{15} + \frac{8}{5\sqrt{3}} + \frac{4}{3}\log\sqrt{3} + \frac{2}{3\sqrt{3}}\big[\mathrm{Li}_2(d) - \mathrm{Li}_2(-d)\big].$$

Substitution in (3) gives

$$U_{\text{rad}} = \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9}\log\sqrt{3} + \frac{2}{9\sqrt{3}}\left[\mathrm{Li}_2(2-\sqrt{3}) - \mathrm{Li}_2(\sqrt{3}-2)\right].$$

This proves the analytic benchmark $\mathrm{OPT} \le U_{\text{rad}}$. We next improve it by an exact divergence-free correction.

7.1 A rational continuous stream certificate

For one bidder with own type (x, y), opponent type (z, w), and $r = \max(x, y)$, write the radial vector field as

$$F^0(x, y) = \frac{3r^2 - 1}{2r^2}(x, y).$$

Define $F^0(0,0) = 0$; this value is immaterial. Although the field is singular at the origin, $|F^0| = O(1/r)$, so it is locally integrable in two dimensions. Its classical divergence away from the origin is 3, its outer boundary flux is 2, and its inward flux around a puncture at the boundary corner $(0,0)$ tends to 1. Because that corner is not an interior point of the open square, we use the following punctured-domain identity rather than a bare distributional-divergence shorthand. Applying Green's formula outside the square $[0,\varepsilon]^2$ and then letting $\varepsilon \downarrow 0$ gives, for every Lipschitz truthful utility,

$$\int F^0 \cdot \nabla u = \int_{x=1} u + \int_{y=1} u - 3\int_{[0,1]^2} u + u(0).$$

Indeed, the puncture flux tends to one and Lipschitz continuity replaces the trace of u there by $u(0)$ with vanishing error. This is equivalently the integration-by-parts identity

$$\int_{[0,1]^2} p(v)\, dv = \int_{[0,1]^2} F^0(v) \cdot X(v)\, dv - u(0).$$

We now add an opponent-dependent curl. For an exponent $e = (e_0, e_1, e_2, e_3)$, put

$$m_e = x^{e_0} y^{e_1} z^{e_2} w^{e_3}, \qquad \sigma(e) = (e_1, e_0, e_3, e_2).$$

Let E be the lexicographically smaller member of every nontrivial σ-pair of total degree at most four. Define

$$h(x, y, z, w) = \sum_{e \in E} \theta_e (m_e - m_{\sigma(e)}), \quad \psi = x(1-x)y(1-y)h, \quad G = (\partial_y \psi, -\partial_x \psi).$$

The thirty-two rational coefficients are listed below in the exact basis order used by the verifier. A convex quasi-Monte Carlo search was used only to discover them; the proof defines the field by the following rational numbers and does not trust the numerical optimization.

e	θ_e	e	θ_e
$(0,0,0,1)$	$-2995385/993184$	$(0,0,0,2)$	$2337131/676164$
$(0,0,0,3)$	$5724621/964727$	$(0,0,0,4)$	$-4591777/714341$
$(0,0,1,2)$	$-834567/846478$	$(0,0,1,3)$	$6291553/970203$
$(0,1,0,0)$	$2762675/446201$	$(0,1,0,1)$	$206964/961681$
$(0,1,0,2)$	$-6090207/979156$	$(0,1,0,3)$	$6862090/913753$
$(0,1,1,0)$	$-5206994/854575$	$(0,1,1,1)$	$515889/117356$
$(0,1,1,2)$	$-3346363/843816$	$(0,1,2,0)$	$-125263/465506$
$(0,1,2,1)$	$3005651/895053$	$(0,1,3,0)$	$-1043238/468269$
$(0,2,0,0)$	$1660919/994155$	$(0,2,0,1)$	$1276309/943222$
$(0,2,0,2)$	$-2075136/255347$	$(0,2,1,0)$	$404816/59269$
$(0,2,1,1)$	$250988/352269$	$(0,2,2,0)$	$3825425/994154$
$(0,3,0,0)$	$-5383501/354845$	$(0,3,0,1)$	$3312081/940771$
$(0,3,1,0)$	$-4885471/759831$	$(0,4,0,0)$	$7648954/974695$
$(1,1,0,1)$	$-603823/508896$	$(1,1,0,2)$	$417332/188831$
$(1,2,0,0)$	$17884893/955196$	$(1,2,0,1)$	$992961/785407$
$(1,2,1,0)$	$2761171/712822$	$(1,3,0,0)$	$-3841999/611370$

For every fixed opponent report, $\operatorname{div}_{x,y} G = 0$. Moreover $G \cdot n = 0$ on each side of the own-type square: on a vertical side $\partial_y \psi = 0$, and on a horizontal side $-\partial_x \psi = 0$. Therefore adding G changes neither the conservation law nor the boundary term. Indeed, the polynomial field is bounded and the Lipschitz utility belongs to $W^{1,\infty}$, so weak Green integration gives $\int G \cdot \nabla u = 0$ on almost every opponent slice; its joint polynomial dependence then permits Fubini. With bidder 2 receiving the same field after swapping (x, y) with (z, w), set $F_i = F_i^0 + G_i$. For every randomized feasible allocation,

$$\sum_i F_{ij} X_{ij} \le \max\{0, F_{1j}, F_{2j}\},$$

and consequently

$$\mathrm{OPT} \le \int_{[0,1]^4} \sum_{j=1}^{2} \max\{0, F_{1j}(v), F_{2j}(v)\}\, dv. \tag{4}$$

No implementability or weak-monotonicity property of the pointwise maximizing allocation is used here.

It remains to bound the integral in (4) exactly. Split each bidder square into (s, st) and (st, s). In each of the four chart pairs, multiplication by the Jacobian $s_1 s_2$ turns both item-1 competitors into exact rational polynomials in (s_1, t_1, s_2, t_2). The antisymmetry of h under

$(x, y, z, w) \mapsto (y, x, w, z)$ exchanges the two field components; invariance of the uniform distribution therefore makes the item-2 integral identical. The two charts overlap only on the diagonal and at the origin, which are null sets.

The certificate converts each polynomial and their exact difference to tensor Bernstein form. A polynomial lies in the convex hull of its Bernstein control coefficients, and its box integral is the box volume times their mean. Every exact initial control c is stored as an integer $m = \lfloor 10^9 c \rfloor$ with error radius one. A degree-n de Casteljau half-box split makes n floor averages, so the rigorous radius update is $E \leftarrow E + n$. Lower bounds for a competitor and for the exact competitor difference certify a fixed winner; otherwise the maximum upper control gives a valid supremum charge.

The base tree has depth 20. Until its final four levels, it splits on the axis of largest adjacent-control variation. In the final four levels it chooses the axis with smallest immediate certified two-child charge. For every box still unresolved at depth 20, the adaptive step compares the hold charge $2b$, expressed at common depth 21, with the sum of the two valid child charges for each of the four possible axes. It splits only when one child sum is strictly smaller. Taking the minimum of several valid upper bounds is valid. Moreover, a fixed box at level ℓ is multiplied by $2^{21-\ell}$, exactly the number of common-depth descendants, so every charge has the correct dyadic volume normalization.

The manifest-driven verifier reconstructs the polynomials and reports

$$\begin{aligned} \text{fixed-winner or zero base boxes}: &\quad 235487, \\ \text{unresolved depth-20 boxes}: &\quad 277676, \\ \text{adaptively split / held boxes}: &\quad 233184 \,/\, 44492, \\ \text{integer accumulator}: &\quad 930318295428931. \end{aligned}$$

After the four chart pairs and the item-symmetry factor two, this gives

$$\int \sum_j \max\{0, F_{1j}, F_{2j}\} \le \frac{2 \cdot 930318295428931}{10^9\, 2^{21}} = \boxed{\frac{930318295428931}{1048576000000000}}.$$

This proves the upper half of Theorem 1. The verifier and its clean transcript are in the following certificate directory:

`certificate/continuous_stream_upper_bound/`

7.2 Reproducibility and independent replay

The publication archive separates the proof inputs from the numerical search that found them. The trusted upper-bound directory contains the rational coefficient manifest, polynomial constructor, exact verifier, arithmetic audit, and a captured clean-run transcript; the root SHA-256 manifest covers these files. The verifier checks all manifest expectations and fails closed on any mismatch. Before fixed-point conversion, every polynomial and Bernstein coefficient is constructed with `fractions.Fraction`. The largest axis degree is eight, so the initial one-unit rounding enclosure grows to at most $1+8\cdot 21 = 169$ fixed-point units. The maximum initial absolute control is 1475346733; the conservative pair-operation bound $2950693802 < 2^{63}-1$ covers every signed 64-bit sum and difference used by subdivision. Control means, shifts, child-charge sums, and the global accumulator use arbitrary-precision Python integers.

A second program in `certificate/independent_stream_upper_bound/` reimplements the basis, chart maps, Bernstein conversion, subdivision, winner tests, and coverage accounting without importing the formal verifier or its polynomial module. Its complete adaptive replay gives the same accumulator, all refinement counts, and complete coverage 8388608/8388608. A separate shallow mode carries the exact scaled rational controls alongside the integer controls and performs 3,087,315 enclosure checks and 444 charge checks. A same-code replay at fixed-point scale 10^8 gives the still-smaller-than-incumbent bound 18606431629403/20971520000000, checking that the improvement is not a 10^9-scale rounding artifact. These replays audit the finite arithmetic; the surrounding continuous weak-duality argument is the analytic proof above.

The lower-bound directories similarly reconstruct the 144 affine/pivot cells and the surcharge polynomial from rational manifests. The root reproduction command reruns all publication-facing checks, compares their structured outputs with the theorem endpoints, and verifies the archive hash manifest. No optimizer output is read anywhere on the trusted proof path.

8 Discussion and open problem

The certified lower and upper values are unequal. The corrected stream bound still relaxes cross-type convex-gradient compatibility after using DSIC, so equality would require a pointwise field maximizer that need not itself be cyclically monotone. Conversely, the lower mechanism is one explicit deterministic menu construction; no valid theorem reduces the allowed class of randomized mechanisms to this family. Consequently neither the mechanism nor its revenue is asserted to be globally optimal.

The stream certificate is an exact continuous weak-duality witness and narrows the previously reported rigorous bracket, but it is not complementary to the present primal mechanism. Still higher-degree numerical witnesses have sampled envelope values near 0.88325, yet their finite Bernstein bounds deteriorate near winner-switching surfaces. Those values are discovery evidence, not theorem endpoints. A useful next step is therefore to couple witness design to a partition or certificate whose local geometry resolves those surfaces, or to retain more of the convex-gradient restrictions discarded by the itemwise pointwise maximum. Closing the exact primal–dual equality remains open.

Data, code, and materials availability

The accompanying archive contains all exact certificate data, deterministic verification code, transcripts, environment information, and SHA-256 manifests needed to reproduce the stated bounds. Discovery-only experiments are segregated from the trusted proof artifacts.

AI-assisted tools declaration

This manuscript was completed with the assistance of OpenAI GPT-5.6 Sol.

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